MODULE 1 · LECTURE 13

IT Applications for Crop Water and Nutrient Requirements

How computers estimate how much water and fertiliser a crop needs — FAO-56 evapotranspiration, crop coefficients and soil-water budgeting, soil-test-based and targeted-yield fertiliser recommendations — and the software, databases and sensors that put these calculations to work.

Learning objectives

After this lecture you should be able to:

1. Explain reference evapotranspiration, crop coefficients and crop evapotranspiration.
2. Calculate ET_0 with the FAO Penman–Monteith and Hargreaves equations.
3. Calculate seasonal crop water requirement and net and gross irrigation requirements.
4. Use total and readily available water to schedule irrigation.
5. Describe IT tools for crop water management: CROPWAT, ETo Calculator, AquaCrop and WaPOR.
6. Explain soil-test-based and targeted-yield (STCR) fertiliser recommendations and convert nutrient doses into fertiliser products.
7. Describe IT tools for nutrient management: Soil Health Card, Nutrient Expert, leaf colour chart and SPAD meter.

Scope: Lecture 11 introduced models and weather analysis, and Lecture 12 the controllers that deliver water and fertiliser. This lecture shows how the required amounts are calculated. Decision support and expert systems follow in Lectures 16–17, and remote sensing in Lecture 15.

1. Why use IT for water and nutrient requirements?

Deciding how much water and fertiliser a crop needs requires many calculations: daily weather data, crop stages, soil properties, soil-test values and fertiliser contents, repeated for every field and season. Computers do these calculations quickly and consistently, store the data, and pass the results to farmers, advisers and machines. The same calculations sit inside spreadsheets (Lecture 5), crop models (Lecture 11), irrigation and fertigation controllers (Lecture 12) and farm apps (Lecture 14).

2. Crop water requirement: key concepts

Term	Meaning
Evapotranspiration (ET)	Water lost from a field by evaporation from the soil and transpiration from the plants
Reference evapotranspiration (ET_0)	ET from a hypothetical reference grass surface, well watered and actively growing; it expresses the evaporating power of the weather alone
Crop coefficient (Kc)	Ratio of crop ET to ET_0; it depends on the crop and its growth stage
Crop evapotranspiration (ETc)	Water used by a healthy, well-watered crop: ETc = Kc × ET_0 — the crop water requirement
Effective rainfall (Pe)	Part of the rainfall that is stored in the root zone and used by the crop
Net irrigation requirement (NIR)	Water that irrigation must supply: NIR = ETc - Pe
Gross irrigation requirement (GIR)	Water to be applied, allowing for losses: GIR = NIR ÷ application efficiency

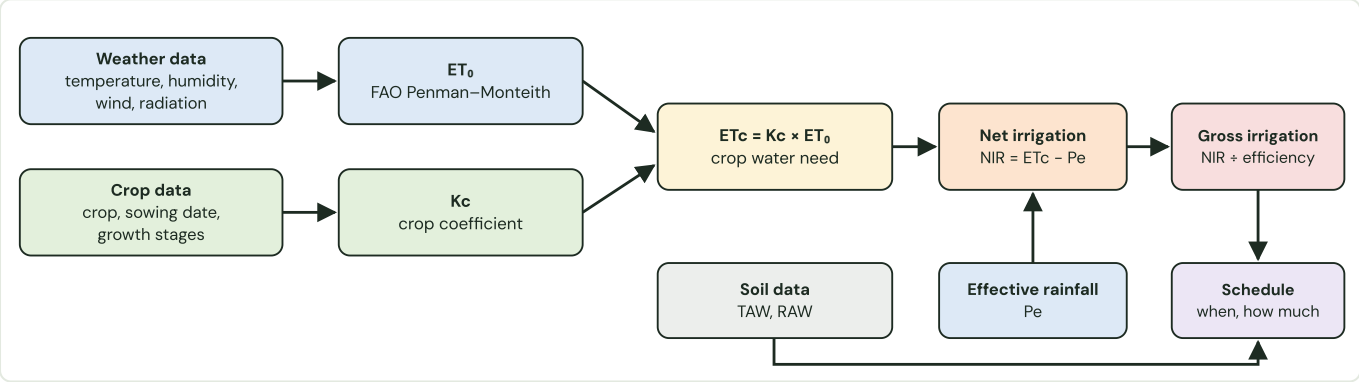


Figure 1. From weather, crop and soil data to an irrigation schedule. Original diagram based on the FAO-56 procedure.

These definitions and procedures follow FAO Irrigation and Drainage Paper 56, the international standard for computing crop water requirements (Allen et al., 1998).

3. Estimating reference evapotranspiration (ETo)

FAO-56 recommends the **FAO Penman–Monteith** equation as the sole standard method for ET_0. It combines the energy available for evaporation (net radiation) with the drying power of the air (vapour pressure deficit and wind) for a reference grass 0.12 m tall (Allen et al., 1998):

$$ET_0 = [\ 0.408\ \Delta\ (Rn - G) + \gamma\ (900\ /\ (T + 273))\ u_2\ (es - ea)\ ] \div [\ \Delta + \gamma\ (1 + 0.34\ u_2)\ ]$$

Here Rn is net radiation and G soil heat flux (MJ m^{-2} day^{-1}); T is mean air temperature (°C); u_2 is wind speed at 2 m (m/s); es - ea is the vapour pressure deficit (kPa); Δ is the slope of the vapour pressure curve and γ the psychrometric constant (kPa/°C).

Method	Data needed	Use
FAO Penman–Monteith	Maximum and minimum temperature, humidity, wind speed, sunshine hours or solar radiation	Standard method whenever full weather data are available
Hargreaves	Maximum and minimum temperature, and extraterrestrial radiation (Ra) from latitude and date	When only temperature data exist; should be checked against Penman–Monteith in each new region (Allen et al., 1998; Hargreaves & Samani, 1985)
Pan evaporation	Daily evaporation from a standard pan × pan coefficient	Simple field method; needs a well-maintained pan

$$\text{Hargreaves: } ET_0 = 0.0023\ (T_{mean} + 17.8)\ \sqrt{(T_{max} - T_{min})} \times Ra \qquad \text{(mm/day)}$$

Worked example 1 — ET_0 for one day (illustrative data, a February day at 400 m altitude, latitude 26.9° N)
Tmax = 26 °C, Tmin = 10 °C (Tmean = 18 °C); RHmax = 80%, RHmin = 35%; u_2 = 2.0 m/s; Rn = 9.0 MJ m^{-2} day^{-1}; G = 0.
Air pressure P = 96.66 kPa → γ = 0.0643 kPa/°C. e°(26) = 3.361 kPa, e°(10) = 1.228 kPa → es = 2.295 kPa; ea = 1.079 kPa; es - ea = 1.215 kPa. Δ at 18 °C = 0.1298 kPa/°C.
Radiation term = 0.408 × 0.1298 × 9.0 = 0.477; aerodynamic term = 0.0643 × (900 / 291) × 2.0 × 1.215 = 0.483.
Denominator = 0.1298 + 0.0643 × (1 + 0.34 × 2.0) = 0.2378 → **ET_0 = (0.477 + 0.483) ÷ 0.2378 = 4.04 mm/day.**
Hargreaves check: Ra = 27.39 MJ m^{-2} day^{-1} = 11.17 mm/day → ET_0 = 0.0023 × (18 + 17.8) × √16 × 11.17 = 3.68 mm/day — close, but lower, because it ignores the dry air and wind.

The same calculation as a Python program (Lecture 8)

```
import math

def sat_vp(t):                        # saturation vapour pressure (kPa) at t (deg C)
    return 0.6108 * math.exp(17.27 * t / (t + 237.3))

def et0_pm(tmax, tmin, rhmax, rhmin, u2, rn, altitude, g=0.0):
    """Daily reference evapotranspiration (mm/day), FAO-56 Penman-Monteith."""
    tmean = (tmax + tmin) / 2
    p = 101.3 * ((293 - 0.0065 * altitude) / 293) ** 5.26     # air pressure, kPa
    gamma = 0.000665 * p                                        # psychrometric constant
    es = (sat_vp(tmax) + sat_vp(tmin)) / 2
    ea = (sat_vp(tmin) * rhmax / 100 + sat_vp(tmax) * rhmin / 100) / 2
    delta = 4098 * sat_vp(tmean) / (tmean + 237.3) ** 2
    num = 0.408 * delta * (rn - g) + gamma * 900 / (tmean + 273) * u2 * (es - ea)
    return num / (delta + gamma * (1 + 0.34 * u2))

et0 = et0_pm(tmax=26, tmin=10, rhmax=80, rhmin=35, u2=2.0, rn=9.0, altitude=400)
kc = 1.15                                                       # wheat, mid-season
print(f"ET0 = {et0:.2f} mm/day; ETc = {kc * et0:.2f} mm/day")

Output:  ET0 = 4.04 mm/day; ETc = 4.64 mm/day
```

The program was run and the output shown is its actual output. Rn was given directly to keep the example short; in practice it is computed from sunshine hours or solar radiation, as programs such as the ETo Calculator do.

4. Crop coefficient and crop water requirement

The crop coefficient changes as the crop grows. FAO-56 divides the season into four stages: **initial** (sowing to about 10% ground cover), **crop development**, **mid-season** (full cover to start of maturity) and **late season** (to harvest). It tabulates typical stage lengths and Kc values for many crops (Allen et al., 1998).

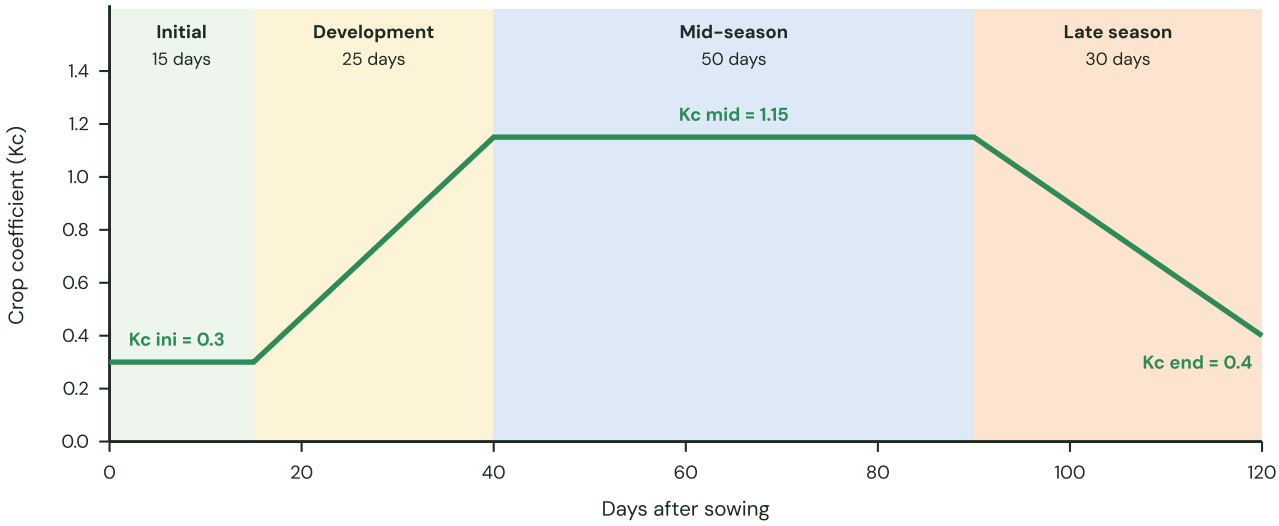


Figure 2. Crop coefficient curve for wheat sown in November in central India — stage lengths and Kc values from FAO-56 Tables 11 and 12. Original chart.

Worked example 2 — seasonal water requirement of wheat (stage lengths and Kc from FAO-56; ET_0 values illustrative)
Initial: 15 days × Kc 0.30 × ET_0 3.4 = 15.3 mm
Development: 25 days × average Kc 0.725 × ET_0 2.8 = 50.7 mm
Mid-season: 50 days × Kc 1.15 × ET_0 3.2 = 184.0 mm
Late season: 30 days × average Kc 0.775 × ET_0 4.8 = 111.6 mm
Seasonal ETc = 361.6 mm. With 20 mm effective rainfall, NIR = 341.6 mm; with 70% surface-irrigation efficiency, GIR = 341.6 ÷ 0.70 = 488 mm, or about 4,880 m³ per hectare.

FAO-56 Kc values are for a sub-humid climate with moderate wind; FAO-56 gives corrections for drier or windier conditions, and Kc ini also depends on how often the soil surface is wetted. Local values from research stations should be used where available.

5. Soil water balance and irrigation scheduling

Irrigation scheduling answers two questions: **when** to irrigate and **how much** to apply. The soil acts as a reservoir. Total available water (TAW) is the water held between field capacity and wilting point in the root zone, but crops start to suffer stress before all of it is used; the fraction p that can be used without stress gives the readily available water (RAW) (Allen et al., 1998).

$$TAW = 1000\,(\theta FC - \theta WP)\,Zr \qquad RAW = p \times TAW \qquad \text{Irrigate when depletion} \approx RAW\text{; net depth} = \text{depletion}$$

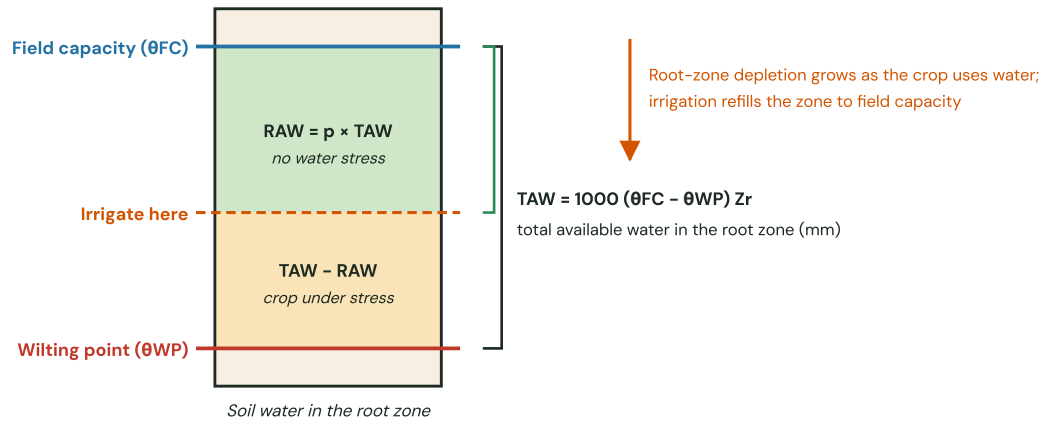


Figure 3. Soil water in the root zone: total and readily available water. Original diagram.

Worked example 3 — irrigation interval for wheat at mid-season (soil values illustrative; p and root depth from FAO-56)
Loam: θFC = 0.30, θWP = 0.14 m³/m³; root depth Zr = 1.0 m; p = 0.55 for wheat.
TAW = 1000 × (0.30 − 0.14) × 1.0 = 160 mm; RAW = 0.55 × 160 = 88 mm.
Mid-season ETc = 1.15 × 3.2 = 3.68 mm/day → interval = 88 ÷ 3.68 ≈ 24 days without rain.
Net depth = 88 mm; gross depth at 70% efficiency = 88 ÷ 0.70 = 126 mm.

Scheduling approach	How the decision is made	IT support
Calendar or crop stage	Fixed dates or critical stages	Crop calendars (Lecture 18)
Water balance (ET-based)	Daily ETc and rain update the depletion; irrigate when it reaches RAW	CROPWAT, spreadsheets, apps using weather data
Soil-moisture sensing	Irrigate when the measured moisture falls below a threshold	Sensors and controllers (Lecture 12)
Weather-based advisories	Advice issued with forecasts	GKMS agromet advisories (Lecture 11)
Remote sensing	Satellite-based evapotranspiration and crop condition	WaPOR database (Lecture 15)

6. IT tools for crop water management

Tool	What it does
CROPWAT 8.0 (FAO)	A decision-support program that calculates crop water and irrigation requirements from climate, crop and soil data, develops irrigation schedules from a daily soil-water balance, and calculates scheme water supply for cropping patterns of up to 20 crops. It uses the methods of FAO-56 and FAO-33, and climate data for over 5,000 stations are available from the companion CLIMWAT database (Food and Agriculture Organization of the United Nations [FAO], n.d.-a)
ETo Calculator (FAO)	Calculates ET_0 from meteorological data by the FAO Penman–Monteith equation (FAO, n.d.-b)
AquaCrop (FAO)	Simulates yield response to water under rainfed and irrigated conditions (Lecture 11)
WaPOR (FAO)	A publicly accessible, near-real-time database based on satellite data for monitoring agricultural water productivity at different scales (FAO, n.d.-c)
Spreadsheets and programs	Custom ET_0, water-balance and scheduling calculations (Lectures 5 and 8)
Controllers and apps	Automatic irrigation from sensors or weather data; scheduling advice on phones (Lectures 12 and 14)

7. Nutrient requirements

A crop takes nutrients from the soil reserve, from organic sources and from fertilisers. The fertiliser needed is the crop's requirement for a given yield minus what the soil and other sources supply, allowing for the fact that only part of the applied fertiliser is taken up. IT helps at every step: storing soil-test data, applying recommendation equations, converting nutrients into fertiliser products, and monitoring the crop.

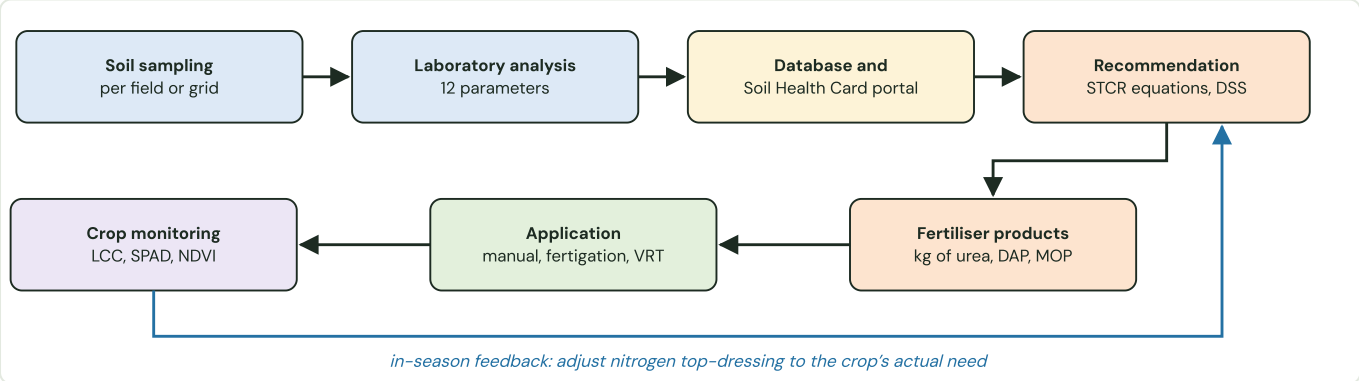


Figure 4. Information flow in soil-test-based nutrient management. Original diagram.

Approach	Basis	Remarks
Blanket (general) recommendation	One dose for a crop in a region	Simple, but ignores differences between fields
Soil-test-based recommendation	Dose adjusted to the soil-test rating (low, medium, high)	Used on the Soil Health Card
Targeted yield (STCR)	Equations relating fertiliser dose to the yield target and soil-test values	Developed by the ICAR All India Coordinated Research Project on Soil Test Crop Response
Plant-based diagnosis	Leaf colour or chlorophyll readings during the season	Adjusts nitrogen top-dressing to the crop's actual need
Decision-support software	Site information, yield goal and nutrient balances	e.g. Nutrient Expert

Soil Health Card

The Soil Health Card reports the status of a farmer's soil for **12 parameters** — nitrogen, phosphorus and potassium (macronutrients), sulphur (secondary nutrient), zinc, iron, copper, manganese and boron (micronutrients), and pH, electrical conductivity and organic carbon — and gives fertiliser recommendations and soil amendments for the farm (Press Information Bureau [PIB], 2025). The cards are generated from laboratory results stored in an online database (Lecture 6).

Targeted-yield (STCR) equations

The STCR approach uses multiple regression on field experiments to relate crop yield to soil-test values and fertiliser inputs, building on the work of Ramamoorthy and colleagues in 1967 (Dey, n.d.). The resulting **fertiliser prescription equations** have the form:

$$\text{Fertiliser nutrient (kg/ha)} = a \times T - b \times S \qquad (T = \text{yield target, kg/ha; } S = \text{soil-test value, kg/ha})$$

Worked example 4 — STCR dose for soybean (published equations for medium black soils, Raisen, Madhya Pradesh; soil-test values illustrative)
FN = 0.0519 T − 0.48 SN; FP_2O_5 = 0.0520 T − 4.10 SP; FK_2O = 0.039 T − 0.22 SK (Raghav et al., 2019).
Yield target T = 1,500 kg/ha; soil test: SN = 120, SP = 6, SK = 200 kg/ha.
FN = 77.85 − 57.6 = 20.3 kg N/ha; FP_2O_5 = 78.0 − 24.6 = 53.4 kg/ha; FK_2O = 58.5 − 44.0 = 14.5 kg/ha.
Products: DAP = 53.4 ÷ 0.46 = 116 kg (it also supplies 116 × 0.18 = 20.9 kg N, enough for the N dose); MOP = 14.5 ÷ 0.60 = 24 kg.

STCR equations are specific to the crop, soil and region for which they were developed; they should not be used elsewhere.

Converting nutrient doses into fertiliser products

Fertiliser	Nutrient content
Urea	46% N
Diammonium phosphate (DAP)	18% N and 46% P_2O_5
Muriate of potash (MOP)	60% K_2O

> **Worked example 5 — products for a 120 : 60 : 40 kg/ha N : P_2O_5 : K_2O recommendation** (illustrative)
> Phosphorus first: DAP = 60 ÷ 0.46 = 130.4 kg, which also supplies 130.4 × 0.18 = 23.5 kg N.
> Remaining N = 120 − 23.5 = 96.5 kg → urea = 96.5 ÷ 0.46 = 209.8 kg (about 4.7 bags of 45 kg).
> Potassium: MOP = 40 ÷ 0.60 = 66.7 kg.
> A spreadsheet or app repeats this for every field once the recommendation and fertiliser contents are entered.

In-season tools

- **Leaf colour chart (LCC) and SPAD chlorophyll meter** — measure leaf greenness, an indicator of nitrogen status. In north-western India, applying 30 kg N/ha each time the SPAD reading of rice fell below 37.5 used 90 kg N/ha and gave yields equal to 120 kg N/ha applied in three fixed splits (Singh et al., 2002).
- **Nutrient Expert** — decision-support software that builds field-specific fertiliser recommendations for cereal crops from information the farmer or adviser can supply (Pampolino et al., 2012).
- **Optical crop sensors and satellite indices (NDVI)** — map crop vigour to guide variable-rate nitrogen (Lectures 12 and 15).
- **Fertigation controllers** — deliver nutrients with irrigation water in small, frequent doses (Lecture 12).

8. Key takeaways

- Crop water requirement is ETc = Kc × ET_0; ET_0 captures the weather, Kc the crop and its stage (FAO-56).
- FAO Penman–Monteith is the standard ET_0 method; Hargreaves is a temperature-only alternative that must be checked locally.
- Net irrigation = ETc − effective rainfall; gross irrigation = net ÷ efficiency.
- TAW = 1000(θFC − θWP)Zr and RAW = p × TAW tell when to irrigate and how much to apply.
- CROPWAT, the ETo Calculator, AquaCrop and WaPOR are FAO tools that automate water calculations.
- Fertiliser needs come from soil tests, targeted-yield (STCR) equations and crop diagnosis; the Soil Health Card reports 12 soil parameters with recommendations.
- Nutrient doses are converted into fertiliser products using their contents — urea 46% N, DAP 18% N and 46% P_2O_5, MOP 60% K_2O.

9. Examination questions

Short-answer questions (2 marks)

Q1. Define reference evapotranspiration.

ET from a hypothetical, well-watered reference grass surface; it represents the evaporating power of the weather.

Q2. Write the relationship between ETc, Kc and ET_0.

ETc = Kc × ET_0.

Q3. Which method does FAO-56 recommend as the standard for ET_0?

The FAO Penman–Monteith method.

Q4. What data does the Hargreaves method need?

Maximum and minimum temperature and extraterrestrial radiation.

Q5. Name the four crop growth stages used for Kc.

Initial, crop development, mid-season and late season.

Q6. Define TAW and RAW.

TAW = 1000(θFC − θWP)Zr, water between field capacity and wilting point; RAW = p × TAW, the part used without stress.

Q7. Net irrigation requirement is 70 mm and efficiency is 70%. Find the gross requirement.

70 ÷ 0.70 = 100 mm.

Q8. What is CROPWAT?

An FAO program that calculates crop water and irrigation requirements and irrigation schedules from climate, crop and soil data.

Q9. List the 12 parameters on a Soil Health Card.

N, P, K, S, Zn, Fe, Cu, Mn, B, pH, EC and organic carbon.

Q10. Write the general form of an STCR fertiliser prescription equation.

Fertiliser nutrient = a × yield target − b × soil-test value.

Long-answer questions (5–10 marks) — answer outlines

Q1. Explain the FAO-56 procedure for estimating crop water requirement, with a flow diagram.

Figure 1; ET_0, Kc, ETc, effective rainfall, NIR and GIR; definitions table.

Q2. Describe the methods of estimating ET_0 and calculate ET_0 by the FAO Penman–Monteith method for given data.

Penman–Monteith equation and terms; Hargreaves and pan methods; worked example 1 and the Python program.

Q3. Explain the crop coefficient curve and calculate the seasonal water requirement of a crop.

Four stages, Figure 2; worked example 2; corrections to Kc.

Q4. How is irrigation scheduled using the soil water balance? Illustrate with an example.

TAW, RAW, Figure 3; worked example 3; scheduling approaches table.

Q5. Describe the IT tools available for crop water management.

CROPWAT and CLIMWAT, ETo Calculator, AquaCrop, WaPOR, spreadsheets and programs, controllers and apps.

Q6. Discuss approaches and IT tools for estimating crop nutrient requirements.

Approaches table; Soil Health Card; STCR (worked example 4); fertiliser products (worked example 5); LCC, SPAD, Nutrient Expert, sensors; Figure 4.

10. Glossary

Term	Meaning
CLIMWAT	FAO climate database used with CROPWAT
Crop coefficient (Kc)	Ratio of crop evapotranspiration to reference evapotranspiration
CROPWAT	FAO program for crop water and irrigation requirements
Effective rainfall	Rainfall stored in the root zone and used by the crop
ET_0	Reference evapotranspiration
ETc	Crop evapotranspiration — crop water requirement
Field capacity	Soil water content after free drainage has stopped
GIR	Gross irrigation requirement
LCC	Leaf colour chart for judging nitrogen need
NIR	Net irrigation requirement
Penman–Monteith	Combination equation for evapotranspiration using energy and aerodynamic terms
RAW	Readily available water — usable without crop stress
SPAD meter	Hand-held meter that measures leaf chlorophyll (greenness)
STCR	Soil Test Crop Response — targeted-yield fertiliser recommendation approach
TAW	Total available water in the root zone
Vapour pressure deficit	Difference between saturation and actual vapour pressure of the air
WaPOR	FAO satellite-based database of agricultural water productivity
Wilting point	Soil water content at which plants can no longer extract water

References

Allen, R. G., Pereira, L. S., Raes, D., & Smith, M. (1998). *Crop evapotranspiration: Guidelines for computing crop water requirements* (FAO Irrigation and Drainage Paper 56). Food and Agriculture Organization of the United Nations. ISBN 92-5-104219-5. https://www.fao.org/4/x0490e/x0490e00.htm

Dey, P. (n.d.). *Sustainable plant nutrient management strategies for food and nutritional security: Current approaches and future strategies* [Poster]. Global Soil Partnership, Food and Agriculture Organization of the United Nations. Retrieved November 9, 2026, from https://fao.org/fileadmin/user_upload/GSP/GSOIL4N/GSOIL4N-Posters/ID_145.pdf

Food and Agriculture Organization of the United Nations. (n.d.-a). *CROPWAT*. Retrieved November 9, 2026, from https://www.fao.org/land-water/resources/tools/software/cropwat/en

Food and Agriculture Organization of the United Nations. (n.d.-b). *ETo Calculator*. Retrieved November 9, 2026, from https://www.fao.org/land-water/resources/tools/software/eto-calculator/en

Food and Agriculture Organization of the United Nations. (n.d.-c). *WaPOR, remote sensing for water productivity*. Retrieved November 9, 2026, from https://www.fao.org/in-action/remote-sensing-for-water-productivity/en

Hargreaves, G. H., & Samani, Z. A. (1985). Reference crop evapotranspiration from temperature. *Applied Engineering in Agriculture, 1*(2), 96–99. https://doi.org/10.13031/2013.26773

Pampolino, M. F., Witt, C., Pasuquin, J. M., Johnston, A., & Fisher, M. J. (2012). Development approach and evaluation of the Nutrient Expert software for nutrient management in cereal crops. *Computers and Electronics in Agriculture, 88*, 103–110. https://doi.org/10.1016/j.compag.2012.07.007

Press Information Bureau. (2025). *Soil Health Card Scheme* [Factsheet]. Government of India. https://www.pib.gov.in/FactsheetDetails.aspx?Id=148602

Raghav, R. S., Singh, Y. V., Kumar, M., Dey, P., & Dubey, S. (2019). STCR based nutrient management in soybean (*Glycine max*) for higher productivity and profitability. *Indian Journal of Agricultural Sciences, 89*(10), 1660–1663. https://doi.org/10.56093/ijas.v89i10.94603

Singh, B., Singh, Y., Ladha, J. K., Bronson, K. F., Balasubramanian, V., Singh, J., & Khind, C. S. (2002). Chlorophyll meter– and leaf color chart–based nitrogen management for rice and wheat in northwestern India. *Agronomy Journal, 94*(4), 821–829. https://doi.org/10.2134/agronj2002.8210

All diagrams, charts and the program are original. Equations and crop parameters are taken from the cited FAO-56 tables; other worked-example values are illustrative, and every result shown was recalculated.

Practice MCQs

Q1. FAO-56 recommends which method as the standard for reference evapotranspiration?

(a) Hargreaves
(b) Blaney–Criddle
(c) FAO Penman–Monteith
(d) Pan evaporation

Q2. Crop evapotranspiration is calculated as

(a) $ETc = ET_0 \div Kc$
(b) $ETc = Kc \times ET_0$
(c) $ETc = ET_0 - Kc$
(d) $ETc = Kc + ET_0$

Q3. The crop coefficient is highest during the

(a) initial stage
(b) development stage
(c) mid-season stage
(d) late-season stage

Q4. The Hargreaves method requires

(a) wind speed and humidity
(b) temperature and extraterrestrial radiation
(c) pan evaporation only
(d) soil moisture

Q5. For θFC = 0.30, θWP = 0.14 and root depth 1.0 m, total available water is

(a) 16 mm
(b) 44 mm
(c) 160 mm
(d) 440 mm

Q6. Readily available water is calculated as

(a) TAW × p
(b) TAW ÷ p
(c) TAW - ETc
(d) $ET_0 \times Kc$

Q7. If net irrigation is 70 mm and efficiency 70%, gross irrigation is

(a) 49 mm
(b) 70 mm
(c) 100 mm
(d) 140 mm

Q8. CROPWAT 8.0 was developed by

(a) ICAR
(b) FAO
(c) IMD
(d) ISRO

Q9. A Soil Health Card reports soil status for how many parameters?

(a) 6
(b) 9
(c) 12
(d) 15

Q10. In STCR equations, the fertiliser dose depends on

(a) the yield target and soil-test value
(b) rainfall only
(c) crop price only
(d) the farm size only

Q11. Diammonium phosphate (DAP) contains

(a) 46% N
(b) 18% N and 46% P_2O_5
(c) 60% K_2O
(d) 16% P_2O_5 only

Q12. A SPAD meter is used to judge a crop's

(a) potassium status
(b) nitrogen status from leaf greenness
(c) soil pH
(d) water stress from soil moisture

Answers

Q1 — (c) FAO Penman–Monteith
Q2 — (b) $ETc = Kc \times ET_0$
Q3 — (c) mid-season stage
Q4 — (b) temperature and extraterrestrial radiation
Q5 — (c) 160 mm
Q6 — (a) TAW × p
Q7 — (c) 100 mm
Q8 — (b) FAO
Q9 — (c) 12
Q10 — (a) the yield target and soil-test value
Q11 — (b) 18% N and 46% P_2O_5
Q12 — (b) nitrogen status from leaf greenness